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PREDICTION, MEASUREMENT AND SIMULATION OF MINERAL LIBERATION

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This presentation covers some recent developments in the study of mineral liberation. I will show that excellent solution methods, both theoretical and practical, are available to predict, measure and simulate mineral liberation phenomena. These methods have been used successfully to solve impoortant operating problems in the base metal and iron ore industries.

Mineral Liberation

Three main problem areas

The prediction problem

The measurement problem

The simulation problem

On this slide, the three main problem areas associated with mineral liberation are specified. The prediction problem deals with the prediction of the liberation, or lack of it, that can be expected when a large mass of ore is comminuted. The measurement problem deals with the measurement, in the laboratory, of the liberation spectrum in a physical sample of particulate matter. The simulation problem deals with the quantitative calculation of liberation that occurs when a real ore is milled in typical industrial comminution circuits and with the calculation of the effects of imperfect liberation on subsequent concentrating operations. Unfortunately some authors have confused the prediction problem with the measurement problem. I will attempt to separate these three problem types and to define their essential features.

The Prediction Problem

Four methods have been developed

Barbery: Boolean model for texture and convex particles.

Meloy: Textural transform and regular particles.

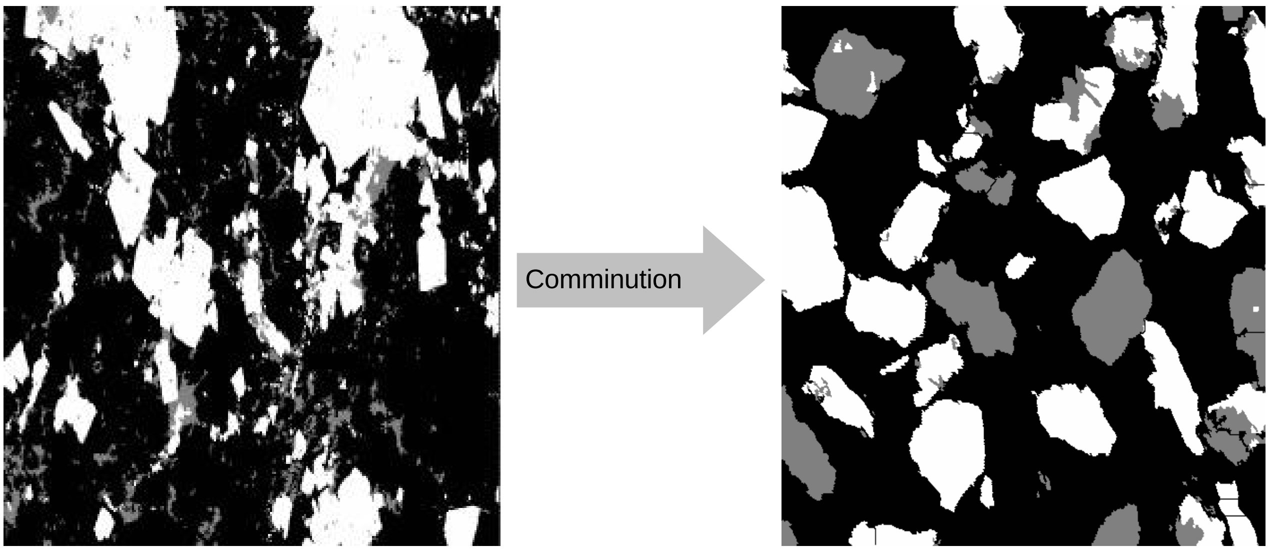
King: Linear intercept sampling of texture and stereological reconstruction.

Simulation: Realistic simulation of texture and fracture pattern.

The prediction problem can be stated as
"Is it possible to characterize the mineralogical texture of an unbroken ore in a quantitative way so that the composition spectrum of the comminuted particles can be predicted as a function of particle size?"
Four methods have been developed to solve this problem. All the methods rely on some quantitaive characterizaton of the mineralogical texture of the ore. This is done nowadays using image analysis usually based on scanning electron microscope imaging.

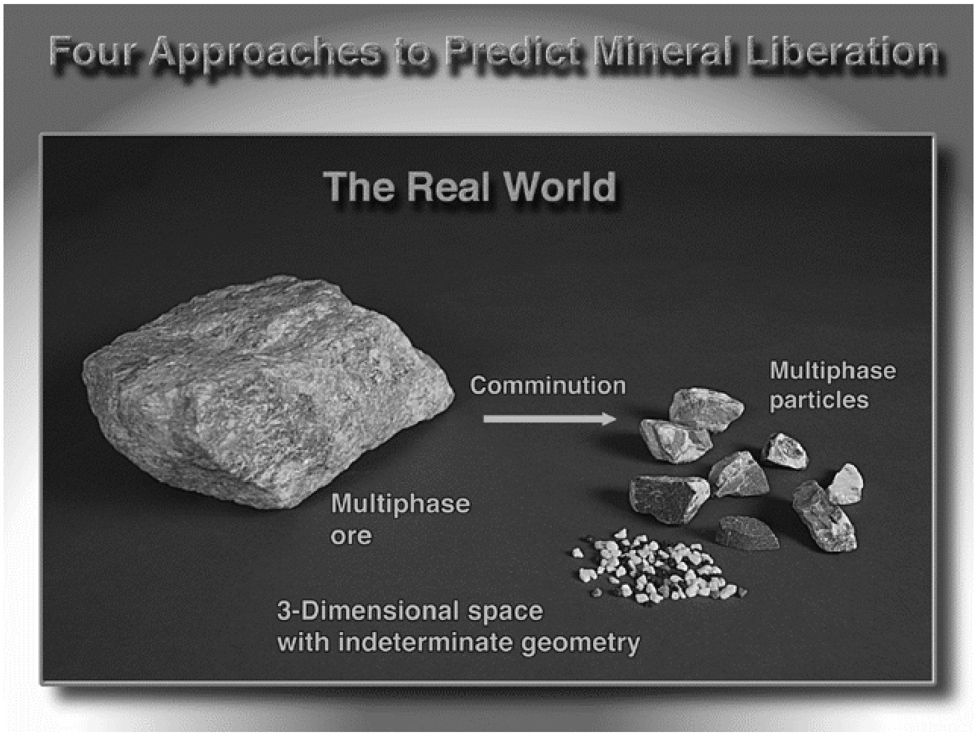
The Prediction Problem

Quantitative characterization of ore texture to predict liberation



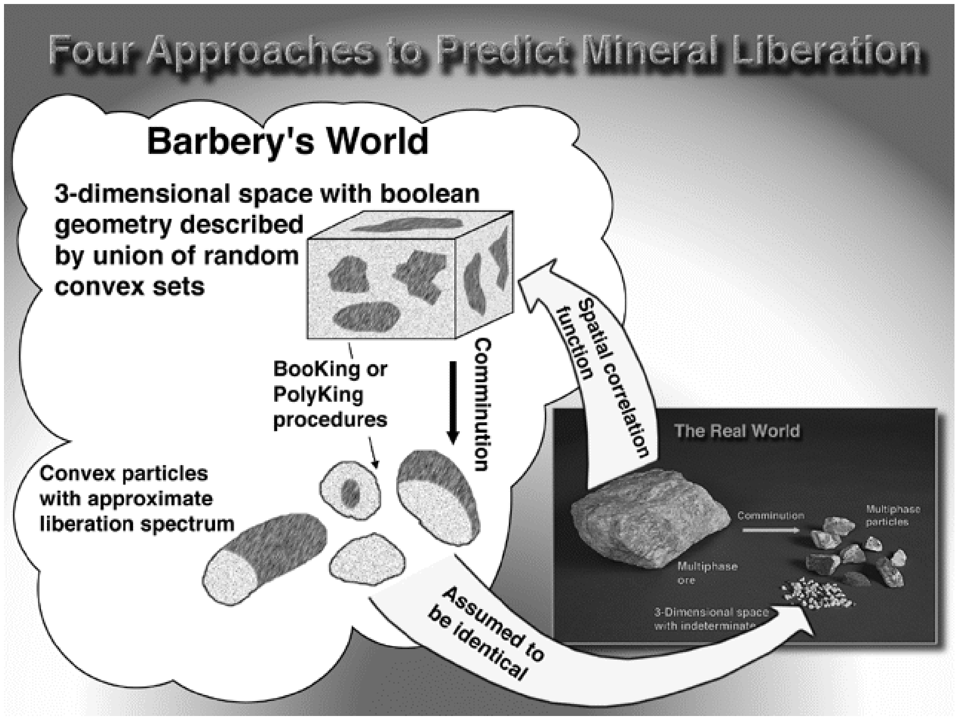
This slide shows a typical mineralogical texture in polished section before comminution on the left and after comminution on the right. This slide defines what the prediction problem is. Is it possible to take images like the one on the left from say a drill core and predict the distribution of valuable minerals over the particle population?

The Prediction problem



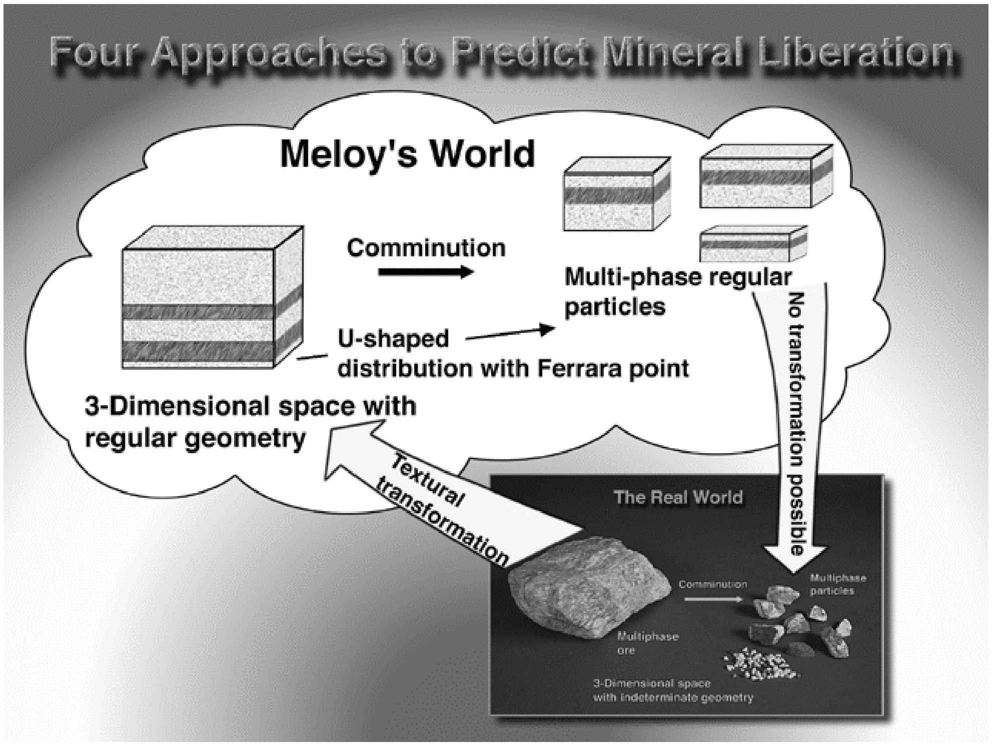
The four methods will be briefly illustrated by means of some graphical representations. Each method has a formal mathematical structure and these are summarized in the associated text. This slide gives a representation of the real-world problem - real rocks containing minerals are comminuted to produce populations of many particles which will be made up of the minerals that are present in the original rock.

The Prediction Problem



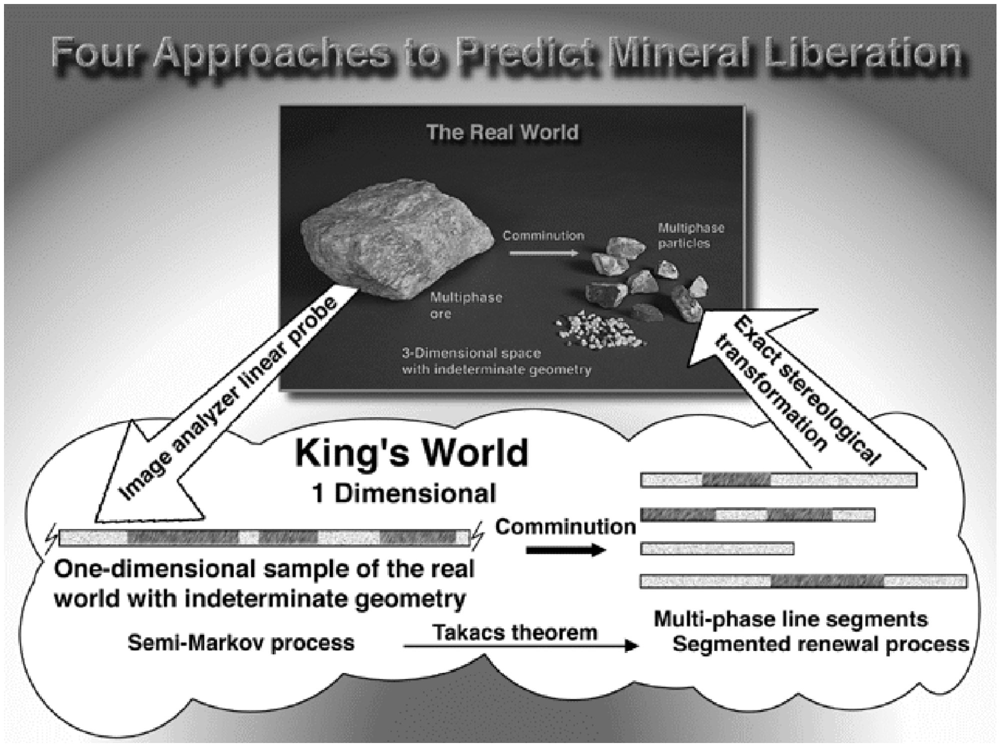
Barbery uses a mathematical description (intersecting random closed sets) to model the actual texture and then predicts the composition distribution among particles that have fairly simple mathematical descriptions. The particles are assumed to be identical to real world particles.

The Prediction Problem



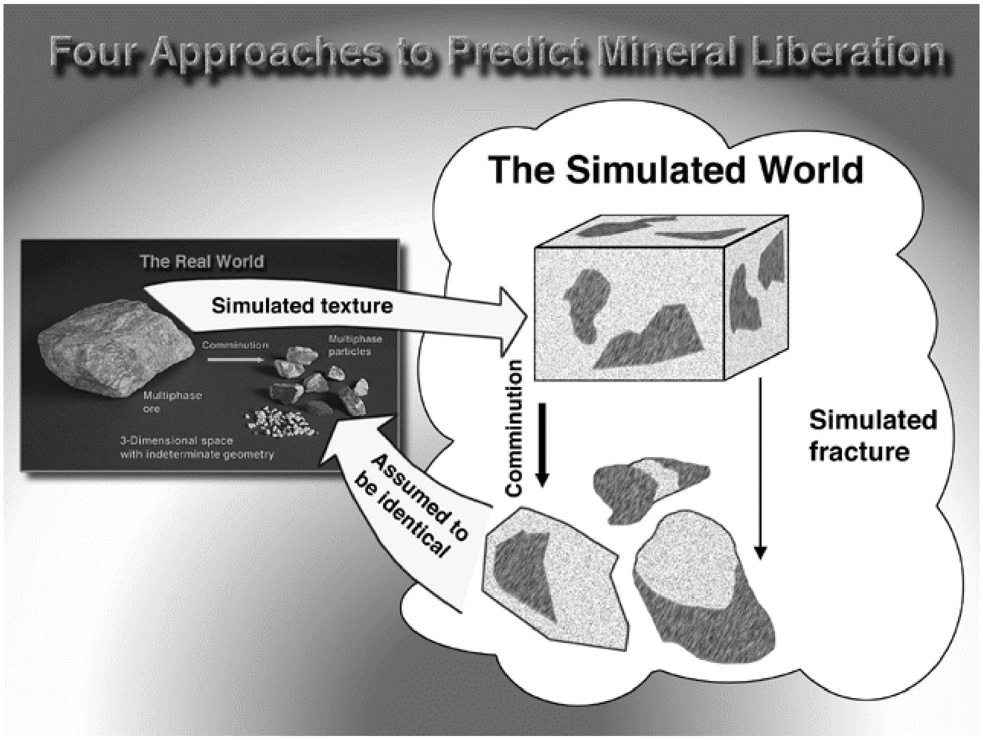
Meloy transforms the mineralogical texture using a textural transformatiom into a simple geometrical form that can be analysed easily after fragmentation using simple geometrical shapes for the particles (cubes,spheres, ellipsoids).

The Prediction Problem



King samples the ore texture using a linear probe and then fractures the probe into linear segments. The 3-dimensional distribution of particle compositions is recovered from the probe fragments by an appropriate stereological transformation.

The Prediction Problem



The simulation method is based on realistic simulations of the ore texture followed by realistic simulations of the comminution operation. The resulting simulated particles are analysed for mineral content.

The Measurement Problem

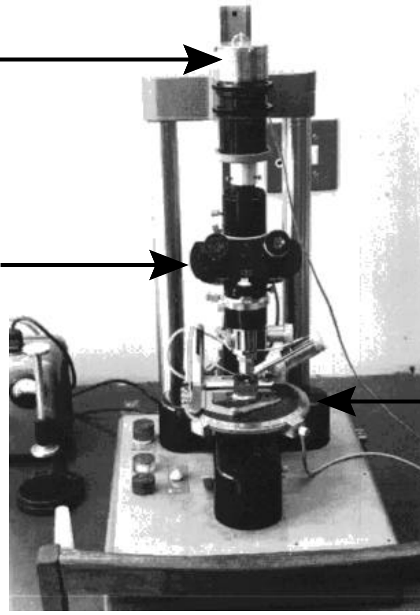
- Particle fractionation
  - Dense liquids or magnetic fluids
- Image analysis
- Stereological correction

Mineral liberation in particulate material can be measured by particle fractionation methods - heavy liquids or magnetic fluids. This is cumbersome, time consuming and expensive.

Image analysis is the effective modern alternative. This method can handle large numbers of samples and is cost effective. The method requires stereological correction before results can be applied effectively to real samples. SEM images collected using back scattered electron dectection are the favored method. These can be acquired quickly and mineral phases that differ by about 0.5 in mass-averaged atomic number can be accurately discriminated.

The Measurement Problem: Image analysis

Early linear scannning image analyser at Wits 1975



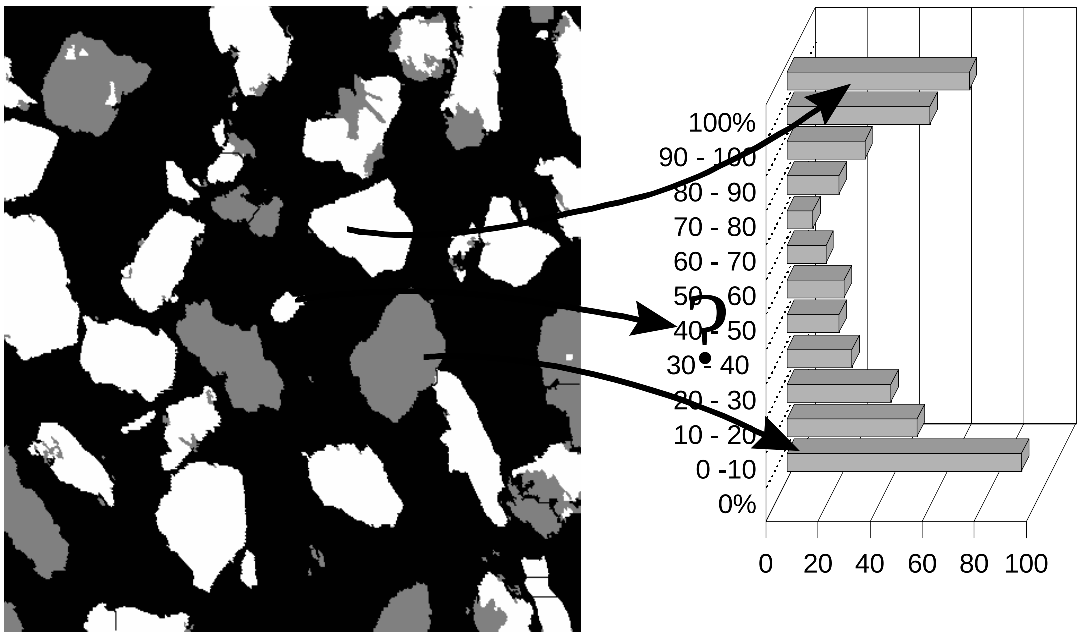
This is a nostalgic picture of my first image analysing microscope which I put together at the University of the Witwatersrand, Johannesburg in 1975. It was used to generate linear intercept distributions in particles and unbroken ore. Because of its mthod of operation it could generate only linear a nd point data. Modern instruments generate areal data as well which is sometimes convenient but not essential.

The Measurement Problem: Image analysis

SEM based image analyser

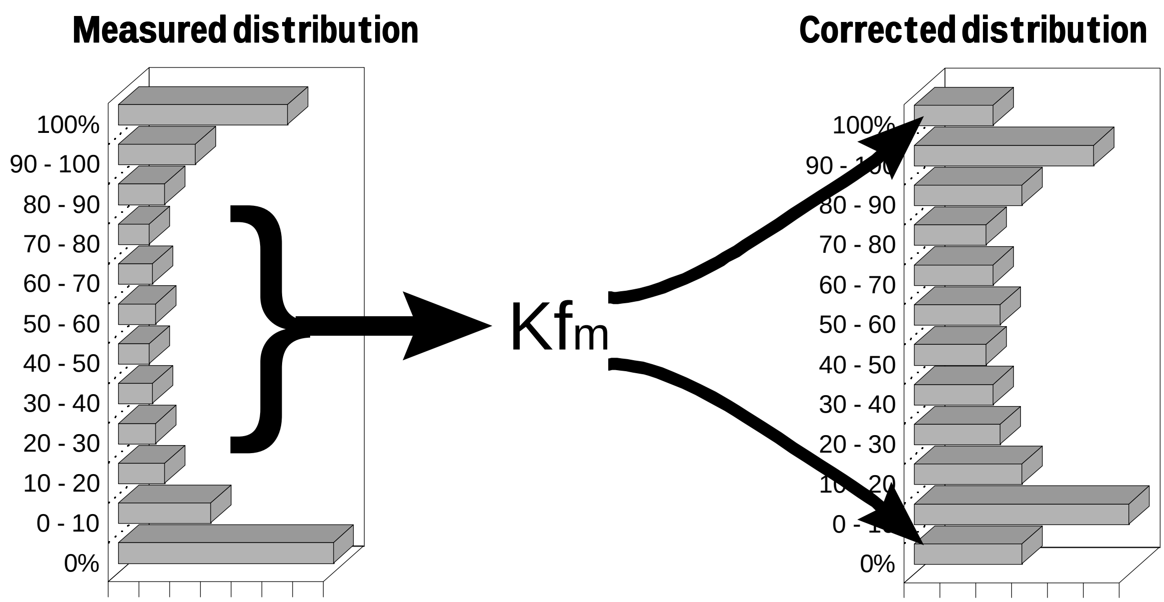
A modern image analysis system at the University of Utah. To use image analysis successfully it is necessary to generate high quality error-free images and to process them using highly specilalized image processing and image analysis techniques. Most of the skill required for successful mineral liberation measurements is contained in the image analysis system that is used. Most general-purpose image processing packages are completely inadequate.

Stereological Correction Methods:
The Allocation Method



The next three slides outline three stereological correction procedures.
This show's Stephen Gay's allocation method for stereological correction. The method seeks to allocate the particles that generate the individual areal features in the image to one of twelve grade bins. Since the true nature of a particle can not be discerned from the section feature, this can be done only on a probalistic basis. For some features the probability of correct allocation is quite high such as for the two large liberated particles that are marked wit red arrows in the slide. However the small feature identified by the blue arrow can be allocated into any one of 11 bins with almost equal probability. The statistical rules for achieving the best allocation are quite complex.

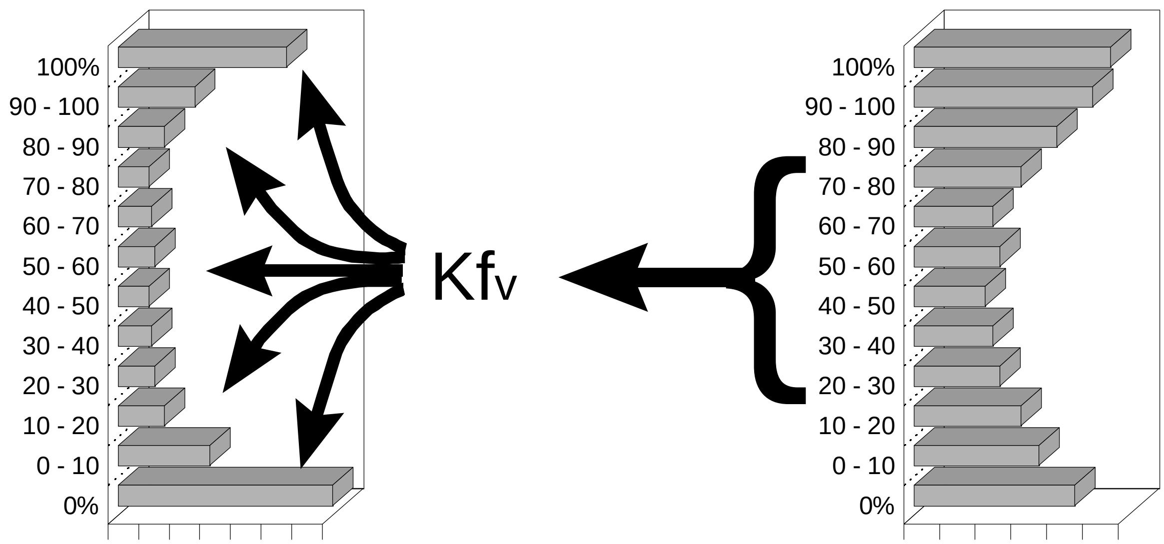
Stereological Correction Methods: Forward correction method.



This slide illustrates the forward correction method developed by Jim Finch's group at McGill University. The stereological transformation matrix is applied to the 10 interior clases of measured data to estimate the level of the stereological error in the two extreme bins. These values are then subtracted from the two extreme bins of the measured areal distribution and the resulting distribution is used as an approximation to the true volumetric distribution.

Stereological Correction Methods: Inverse correction method.

Measured distribution **Corrected distribution**



The inverse correction method developed at the University of Utah uses a common regularization technique that is widely used for the solution of inverse problems. It is necessary to generate an accurate transformation matrix for the texture on hand to use this method successfully. This is the preferred method for rapid and reliable analysis.

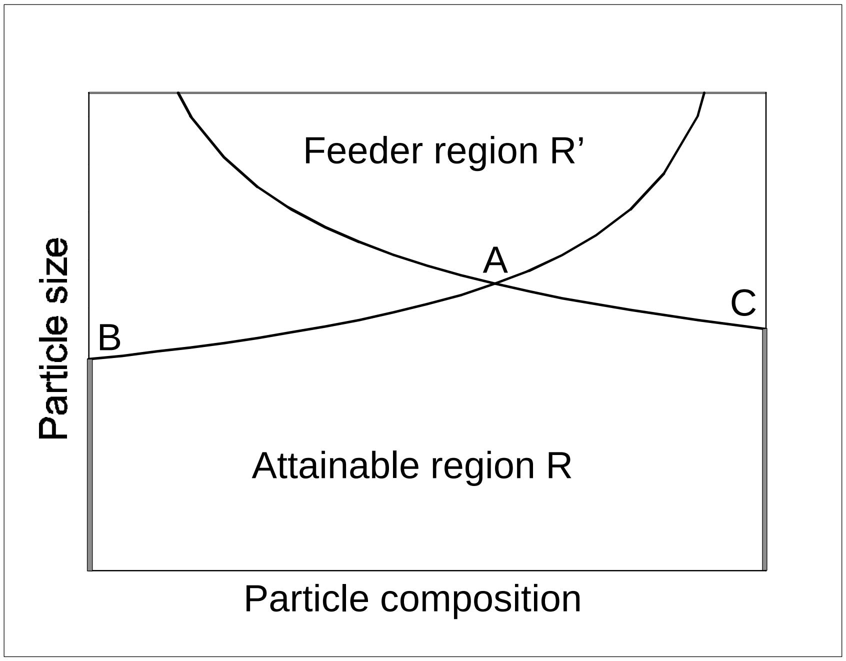
The Simulation Problem

- P Liberation and the population balance method

The most conspicuous success of liberation theory has been the successful simulation of liberation in industrial milling cicuits. The next sequence of slides shows the essence of this method using the Andrews-Mika diagram approach. This method is an extension to the population balance method that is almost universally applied for the simulation of milling circuits and consequently this approach fits easily into existing mineral processing simulation systems.

Andrews-Mika diagram: boundaries

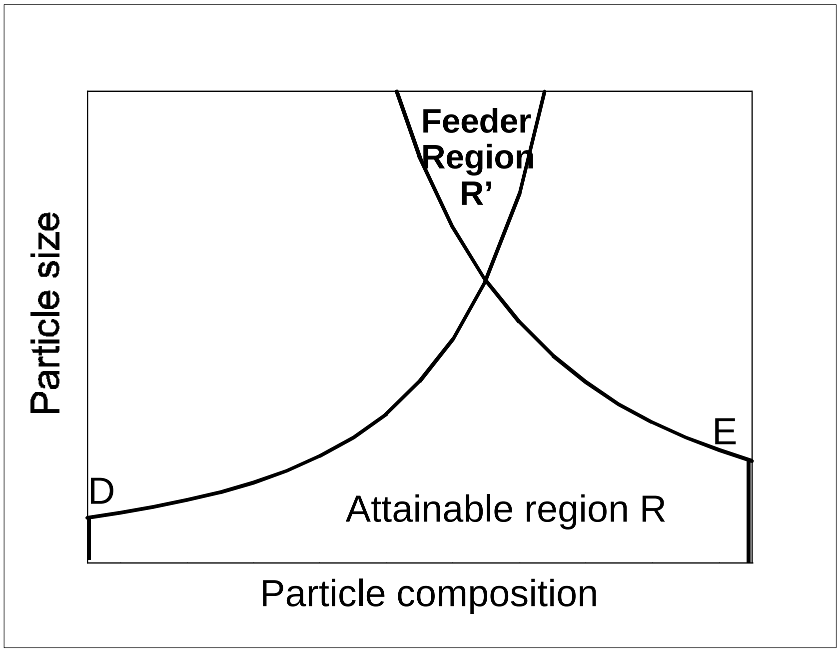
Original form as defined by Andrews and Mika.



The original Andrews-Mika diagram as published by Andrews and Mika. This shows the boundaries of the feeder and attainable regions for a particle located at point A. This particle is a progeny with respect tyo the feeder region and a parent with respect to the attainable region.

Andrews-Mika Diagram: boundaries

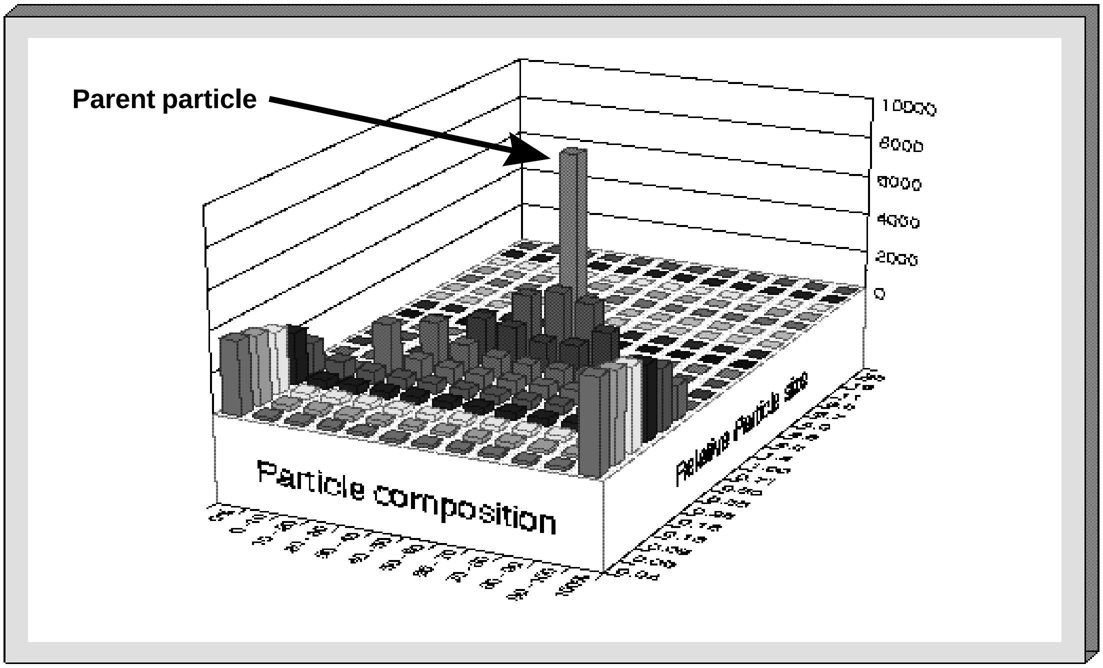
Modified to reflect mineralogical texture.



Early simulations of the breakage of multi-phase particles revealed that the original Andrews-Mika boundaries were too conservative for common ore textures and much narrower boundaries which include textural as well as volume conservation contraints can be used in most cases. This slide shows the first important modification after the PARGEN simulations of the fracture of two-phase particles done at the University of Utah. These boundaries take account of the mineralogical texture as well as the conservation of volume.

Andrews Mika Diagram: internal structure

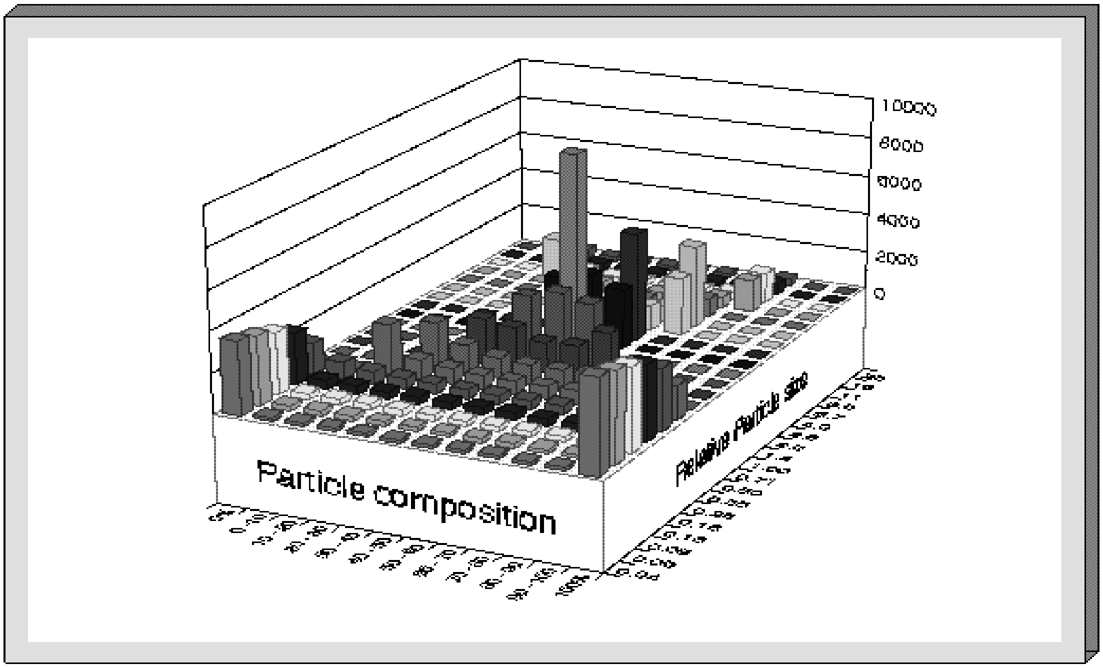
The lower half



The internal structure of the Andrews-Mika diagram must also be modeled before it can be used for simulation of liberation in comminution machines.The lower half (attainable region) of the Andrews-Mika diagram is significantly easier to model than the top half (feeder region)

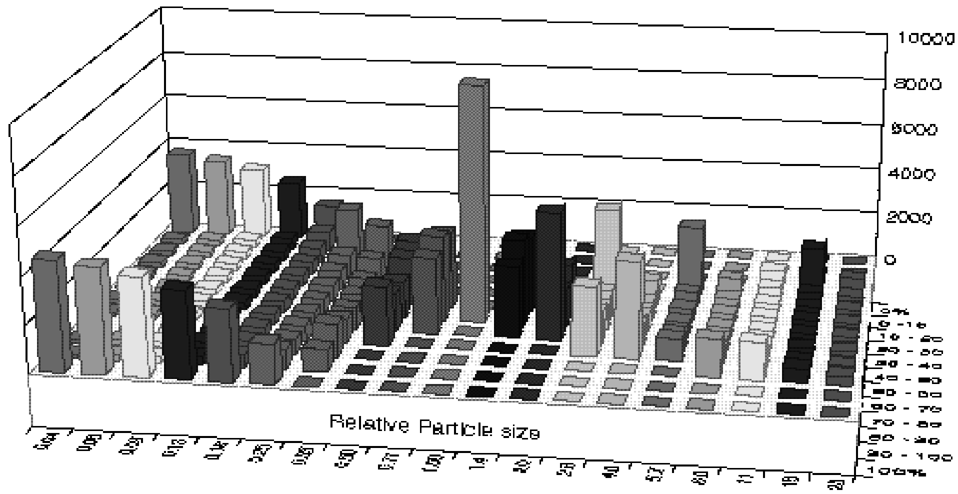
The lower half shows how a parent particle, identified by the arrow, distributes its progeny in the composition particle-size space. However it is the top half of diagram that is most directly useful for simulation work using the population balance method.

Andrews-Mika Diagram: internal structure
Upper half is shown

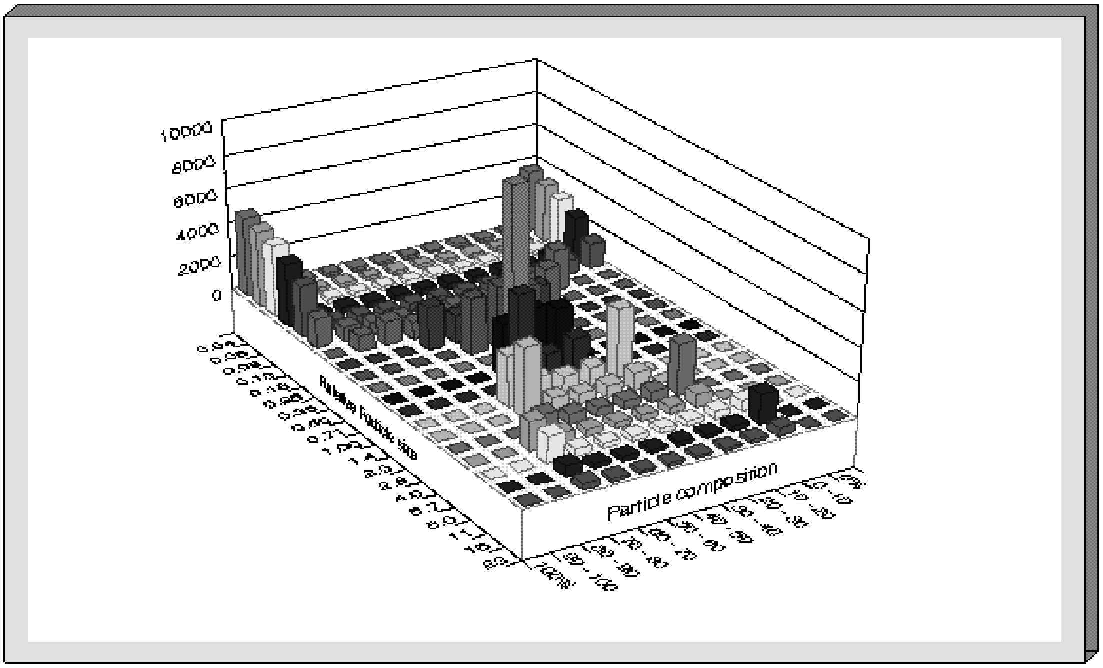


The next three slides show the top half of the Andrews-Mika diagram as the diagram is rotated. The top half can be syntesized from the lower portions of all the A-M diagrams for a particular ore.

Andrews-Mika Diagram: internal structure Rotated

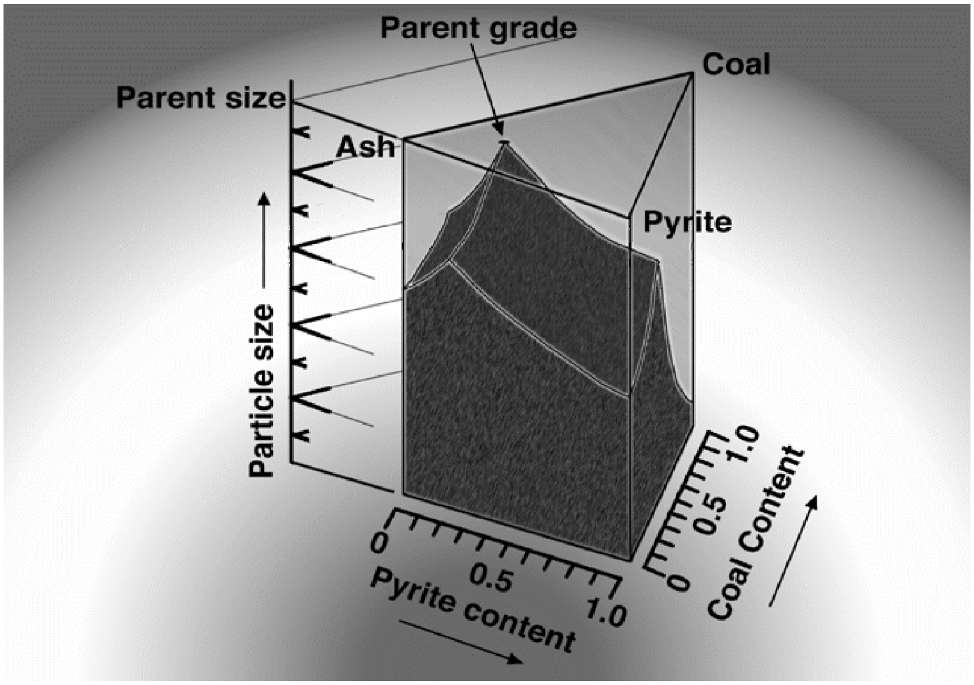


Andrews-Mika Diagram: internal structure
Rotated to show upper half.



Note the lack of easily discernible structure in the top half of the diagram while the lower half has the characteristic bell shape followed by the strong U-shape as the progeny particle size decreases below the parent size. Excellent models for the lower half of the Andrews-Mika diagram have been developed and these have been shown to provide good simulations of the liberation in real ore systems.

Boundaries of the Andrews-Mika diagram for 3-phase systems



The next logical step is the extension of these method to multi-component systems.This slide shows the boundaries of the Andrews-mika diagram for a three-component ore. No models for the boundaries nor the internal structure for these diagrams have yet been developed andthis is a really good area for research at the moment.

SUMMING UP

- P The Prediction Problem:
This has been solved for 2-component systems and promising methods are emerging for multi-component systems.
- P The Measurement Problem:
Image analysis with stereological correction is now applied routinely with excellent results.
- P The Simulation Problem:
Andrews-Mika diagrams provide the necessary information for population balance methods.